

Model Architecture Beats Compute: An Energy-Aware Comparison of Four Transformer Encoders for Financial Sentiment

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Abstract—The amount of energy consumption for deep learning models has come into spotlight as the size of flagship models have grown exponentially. However, it’s unclear if more training energy directly translates into better task performance. This paper addresses that question for transformer-based encoder models when applied to financial sentiment classification. A seven-stage modular pipeline is presented that measures each training and inference run with externally measured CPU and GPU energy usage, sampled at 1Hz and integrated over each run’s time window. Specifically, the paper evaluates four transformer-based encoder models (DistilBERT, BERT-base, RoBERTa-base, and FinBERT) during fine-tuning steps across ten random seeds on 10,521 Coca-Cola (KO) headlines from the FNSPID dataset, with datapoints spanning from 2009 to 2023. One-way ANOVA test across the four models shows significant F1 differences ($F = 18.94$, $p = 1.5 \times 10^{-7}$). An ANCOVA test also reveals that architecture dominates weighted F1 ($p < 10^{-5}$), while training energy provides negligible increases in variance ($p = 0.97$). DistilBERT, the smallest model tested, attains highest energy-adjusted F1 value at roughly half the training energy of larger models. This demonstrates that the architectural choice, rather than the compute budget, drives accuracy on this task.

Index Terms—energy-efficient machine learning, transformer models, financial sentiment analysis, ANCOVA, weak supervision

I. INTRODUCTION

Emerging transformer-based encoder models are demanding greater and greater energy demands for training and deploying. This has prompted calls for the field to evaluate efficiency in addition to accuracy during model decisions [6]. Strubell et al. [7] quantified the carbon footprint of large language model (LLM) training and found that a single high-end model can emit hundreds of times more CO₂ over its lifetime than a typical passenger vehicle. In applied financial natural language processing, professionals typically use large pre-trained transformers (most commonly BERT-base [2], RoBERTa-base [3], or domain-adapted variants such as FinBERT [4]) for short sentiment classification tasks. Oftentimes, this is done without empirical evidence that the additional compute leads to substantial improved task performance. Ultimately, this can worsen returns due to increased latency that leads to trade losses. Smaller, distilled models like DistilBERT [1] are frequently dismissed due to lower observed accuracies, despite

having a much lower energy footprint. The assumption that “bigger is better” persists in financial NLP literature largely by inheritance from general-purpose benchmarks.

Existing work on energy-efficient machine learning focuses on training-time energy measured across general-domain benchmarks such as GLUE or ImageNet [6], [7], with relatively small attention on the inference time-energy cost, which can become significant at scale. This paper addresses that gap by treating architecture as the independent variable, fixing the training protocol across models, and bracketing every training and inference run with externally measured energy at 1 Hz resolution, enabling a direct statistical test of whether energy budget contributes any explanatory power once architecture is partialled out.

This paper has three main contributions. First, it presents a seven-stage pipeline that brackets every training and inference run with 1 Hz CPU and GPU energy logging. It combines Intel RAPL package counters with NVIDIA NVML power telemetry and integrates the resulting traces over ISO 8601 wall-clock windows. Second, it provides an evidence-based comparison of four transformer-based encoder models DistilBERT [1], BERT-base [2], RoBERTa-base [3], and FinBERT [4]. Each model is fine-tuned on weak-labeled financial headlines drawn from the FNSPID corpus [5] for The Coca-Cola Company (KO) over a period from 2009 to 2023. Third, it reports a statistical analysis using analysis of variance and analysis of covariance that demonstrates that architecture choice is a significant determinant of F1 on the task and that training-energy budget alone explains negligible variance in accuracy once architecture is controlled for. Under this analysis, DistilBERT achieves the highest energy-adjusted F1 and happens to be the smallest model.

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